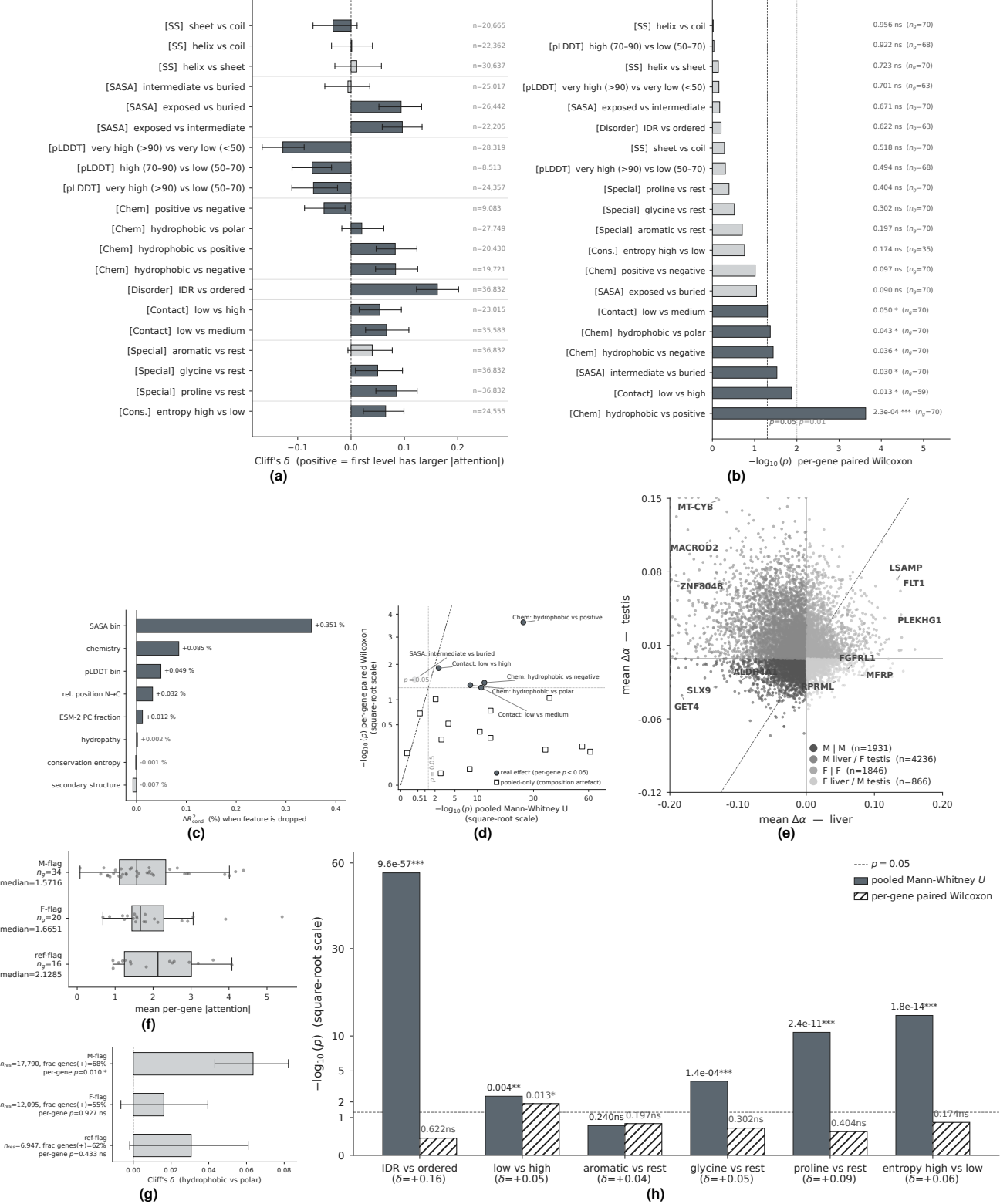

[SS] sheet vs coil
[SS] helix vs coil
[SS] helix vs sheet
[SASA] intermediate vs buried
[SASA] exposed vs buried
[SASA] exposed vs intermediate
[pLDDT] very high (>90) vs very low (<50)
[pLDDT] high (70–90) vs low (50–70)
[pLDDT] very high (>90) vs low (50–70)
[Chem] positive vs negative
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs positive
[Chem] hydrophobic vs negative
[Disorder] IDR vs ordered
[Contact] low vs high
[Contact] low vs medium
[Special] aromatic vs rest
[Special] glycine vs rest
[Special] proline vs rest
[Cons.] entropy high vs low
n=20,665
n=22,362
n=30,637
n=25,017
n=26,442
n=22,205
n=28,319
n=8,513
n=24,357
n=9,083
n=27,749
n=20,430
n=19,721
n=36,832
n=23,015
n=35,583
n=36,832
n=36,832
n=36,832
n=24,555
−0.1 0.0 0.1 0.2
Cliff's δ (positive = first level has larger |attention|)
(a)
[SS] helix vs coil
[pLDDT] high (70–90) vs low (50–70)
[SS] helix vs sheet
[pLDDT] very high (>90) vs very low (<50)
[SASA] exposed vs intermediate
[Disorder] IDR vs ordered
[SS] sheet vs coil
[pLDDT] very high (>90) vs low (50–70)
[Special] proline vs rest
[Special] glycine vs rest
[Special] aromatic vs rest
[Cons.] entropy high vs low
[Chem] positive vs negative
[SASA] exposed vs buried
[Contact] low vs medium
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs negative
[SASA] intermediate vs buried
[Contact] low vs high
[Chem] hydrophobic vs positive
0.956 ns (n_g=70)
0.922 ns (n_g=68)
0.723 ns (n_g=70)
0.701 ns (n_g=63)
0.671 ns (n_g=70)
0.622 ns (n_g=63)
0.518 ns (n_g=70)
0.494 ns (n_g=68)
0.404 ns (n_g=70)
0.302 ns (n_g=70)
0.197 ns (n_g=70)
0.174 ns (n_g=35)
0.097 ns (n_g=70)
0.090 ns (n_g=70)
0.050 * (n_g=70)
0.043 * (n_g=70)
0.036 * (n_g=70)
0.030 * (n_g=70)
0.013 * (n_g=59)
2.3e-04 *** (n_g=70)
p=0.05 p=0.01
0 1 2 3 4 5
$-\log_{10}(p)$ per-gene paired Wilcoxon
(b)
SASA bin
chemistry
pLDDT bin
rel. position N→C
ESM-2 PC fraction
hydropathy
conservation entropy
secondary structure
+0.351 %
+0.085 %
+0.049 %
+0.032 %
+0.012 %
+0.002 %
-0.001 %
-0.007 %
0.0 0.1 0.2 0.3 0.4
ΔR^2_{cond} (%) when feature is dropped
(c)
Chem: hydrophobic vs positive
SASA: intermediate vs buried
Contact: low vs high
Chem: hydrophobic vs negative
Chem: hydrophobic vs polar
Contact: low vs medium
p = 0.057
p = 0.05
real effect (per-gene p < 0.05)
pooled-only (composition artefact)
0 0.5 1 2 3 4
0 0.5 1 2 5 10 30 60
$-\log_{10}(p)$ per-gene paired Wilcoxon (square-root scale)
$-\log_{10}(p)$ pooled Mann-Whitney U (square-root scale)
(d)
MT-CYB
MACROD2
ZNF804B
LSAMP
FLT1
PLEKHG1
FGFRL1
MFRP
ALDH1A1
PRML
SLX9
GET4
M | M (n=1931)
M liver / F testis (n=4236)
F | F (n=1846)
F liver / M testis (n=866)
0.15 0.08 0.01 -0.06 -0.12
-0.20 -0.10 0.00 0.10 0.20
mean Δα — testis
mean Δα — liver
(e)
M-flag
n_g=34
median=1.5716
F-flag
n_g=20
median=1.6651
ref-flag
n_g=16
median=2.1285
0 1 2 3 4 5
mean per-gene |attention|
(f)
M-flag
n_{res}=17,790, frac genes(+)=68%
per-gene p=0.010 *
F-flag
n_{res}=12,095, frac genes(+)=55%
per-gene p=0.927 ns
ref-flag
n_{res}=6,947, frac genes(+)=62%
per-gene p=0.433 ns
0.00 0.02 0.04 0.06 0.08
Cliff's δ (hydrophobic vs polar)
(g)
p = 0.05
pooled Mann-Whitney U
per-gene paired Wilcoxon
$-\log_{10}(p)$ (square-root scale)
60 30 10 5 2 1 0
9.6e-57***
0.622ns
0.004**
0.013*
0.240ns
0.197ns
1.4e-04***
0.302ns
2.4e-11***
0.404ns
1.8e-14***
0.174ns
IDR vs ordered (δ=+0.16)
low vs high (δ=+0.05)
aromatic vs rest (δ=+0.04)
glycine vs rest (δ=+0.05)
proline vs rest (δ=+0.09)
entropy high vs low (δ=+0.06)
(h)